



AIR5011 / MDS5122 / MBI6011

Final Project

Due by: 23:59, May 6 (Wednesday), 2026

Instructions:

1. You must submit your files on Blackboard. Upload a PDF file along with your code (excluding large files such as datasets).
 2. Your submission needs to be clearly-structured and well-presented to receive full credit. Please detail how you leverage your acquired knowledge in class to solve the problem and present your proposed solution step by step. Ensure that your report is written in English.
 3. The programming language is Python, and your code should include necessary comments so that others can easily follow its logic.
 4. Your report should reflect your individuality and originality. If your answers are inspired by other sources (e.g., the Internet or AI), maintain academic integrity by including a reference or acknowledgment in your submitted report to indicate how these sources inspired you.
 5. Plagiarized projects or reports will not be graded. Late submissions are subject to the penalty specified in the class syllabus.
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Autoregressive Speech Prediction (100 points)

With the development of large language models (LLMs) and voice agents, speech is becoming increasingly important in today's speech-text systems. One important task is speech prediction, where a model predicts future speech frames or segments given the previous speech context. This task is very similar to language modeling where previous words/tokens are used to predict the current word. In this final project, you are required to implement, through self-supervised learning, an autoregressive speech prediction model that operates on discrete speech tokens (see Fig. 1 as an example).

Specifically, a system performing such speech prediction has been recently published [1] (see Fig. 1). You can replicate this system, make changes to it, or develop a new system performing the same task. You should use and compare the codecs of [EnCodec](#) [2] and [FACodec](#) [3], which encode a speech signal into discrete tokens. Then, implement a codec language model (LM) for predicting the next speech token. Finally, you will need a decoder in the codec to decode speech tokens into a speech signal for the speech prediction task. The experiments should be conducted on the public WSJCAM0 [4] speech dataset (Download [here](#)). You should evaluate your predicted speech on three quantitative metrics: short-time objective intelligibility (STOI) [5], perceptual evaluation of speech quality (PESQ)

[6], and deep noise suppression mean opinion score (DNSMOS) [7]. You can use [TorchMetrics](https://pytorch.org/torchmetrics/) to compute the scores of these metrics. Summarize your implementation details and compare your results with those in [1].

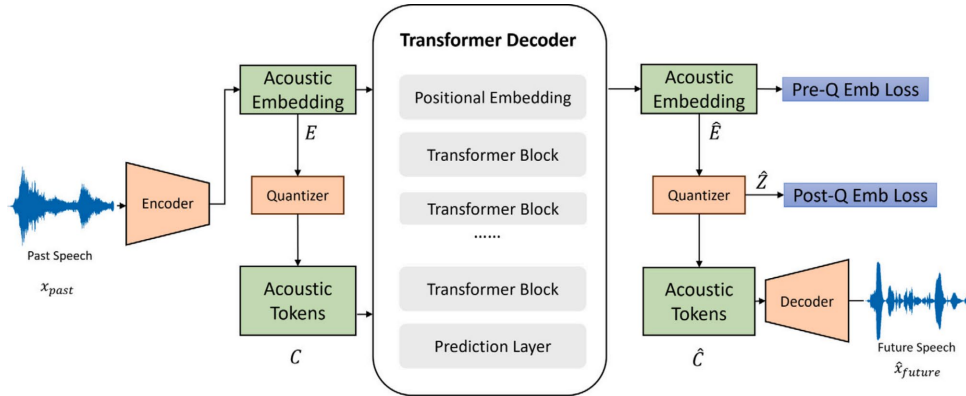


Figure 1. Diagram of a speech prediction model (from [1]).

Your final project report should follow the ICASSP format (see [ICASSP 2026 Author Kit](https://www.icassp.org/2026/author-kit/) for the camera-ready format) with your name, student ID, and affiliation clearly provided. Your report is limited to a maximum of 5 pages (4 pages of content plus one page of references), and Ref. [8] provides an example. You need to upload your code as a supplement to the Blackboard system, along with a Markdown document that clearly instructs the reader on how to set up the environment and reproduce your results. Apart from the requirements stated above, there are no restrictions on how you implement your speech prediction system. The following background and suggestions are provided to assist you in completing this final project.

Background and Suggestions:

1. A speech signal describes continuous vibrations, and its digital (discrete) version gives the amplitude values at regular time points sampled at a certain frequency, e.g., 16 kHz. The discrete version is also called a waveform. Think of it as a long series of samples, one number per sample. The most common representation of a speech signal is the spectrogram, generated by applying short-time Fourier transform (STFT) to a sequence of partly-overlapped time frames of the speech signal. The spectrogram is a 2-dimensional representation of time and frequency, and each time-frequency element contains a magnitude and a phase.
2. A speech codec is a neural network that compresses a waveform signal into discrete latent tokens. The tokens represent compressed speech information, can be modeled like text tokens, and enable autoregressive generation. Each codec token is an integer index of a learned codebook. With speech tokens, the decoder reconstructs the waveform from these

intermediate representations. With the codec, the speech prediction task is transformed into a decoder-only autoregressive LM problem.

3. The training and inference pipeline of your model should look like Fig. 2 in [1] below.

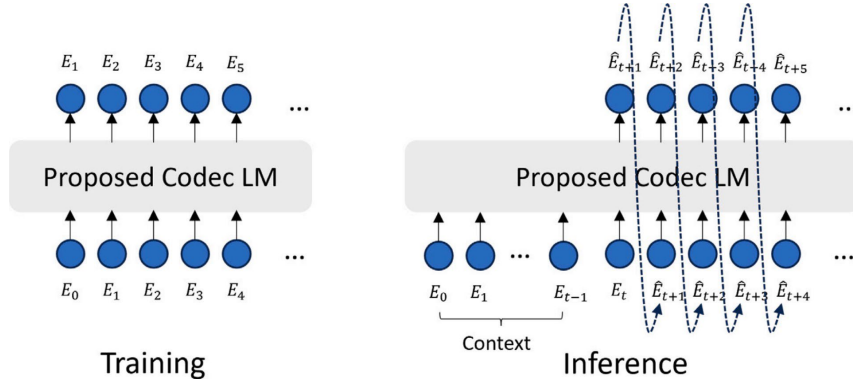


Figure 2. Pipeline for training and inference of a codec LM [1].

4. Evaluation metrics

- STOI (Short-time Objective Intelligibility) measures how understandable predicted speech is to a human listener by computing its short-time spectral correlation with the clean reference signal. STOI values range typically from 0.0 to 1.1, interpreted as a percent-correct score of human speech recognition.
- PESQ (Perceptual Evaluation of Speech Quality) is an ITU-T standardized metric that measures speech quality against clean reference speech by modeling auditory perception. Values range from -0.5 to 4.5, with higher values indicating higher speech quality.
- DNSMOS (Deep Noise Suppression Mean Opinion Score) is a neural network based metric that predicts human-rated speech quality scores without providing a clean reference signal. Values range from 1.0 to 5.0.

Grading Criteria:

Your report must contain the following sections:

1. Introduction (**5 points**)
2. Related work (**5 points**)
3. Method (**30 points**)
4. Experiments (**40 points**)
5. Conclusion & your own remarks (**10 points**)

Quality, clarity, reproducibility, and rationality of the design, code, and report (**10 points**)

Bonus task: Develop a different method to achieve comparable results with those of the best system in [1], or modify the system in [1] to achieve better results (**10 points**).

References:

- [1] H. Wang, Y. Yang, and D.L. Wang, “A speech prediction model based on codec modeling and transformer decoding,” *Computer Speech & Language*, vol. 96, Article 101892, 2026.
- [2] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, “High fidelity neural audio compression,” *Transactions on Machine Learning Research*, 2835-8856, 2023.
- [3] Z. Ju, Y. Wang, K. Shen *et al.*, "NaturalSpeech 3: Zero-shot speech synthesis with factorized codec and diffusion models." in *Proc. ICML*, 2024, pp. 22605-22623.
- [4] J. Garofolo, D. Graff, D. Paul, and D. Pallett, “CSR-I (WSJ0) Complete (LDC93S6A),” *Linguistic Data Consortium*, Philadelphia, 1993.
- [5] C. H. Taal, R. C. Hendriks, R. Heusdens, and J. Jensen, “An algorithm for intelligibility prediction of time–frequency weighted noisy speech,” *IEEE/ACM TASLP*, vol. 19, pp. 2125–2136, 2011.
- [6] A. W. Rix, J. G. Beerends, M. P. Hollier, and A. P. Hekstra, “Perceptual evaluation of speech quality (PESQ)—a new method for speech quality assessment of telephone networks and codecs,” in *Proc. IEEE ICASSP*, 2001, pp. 749–752.
- [7] C.K Reddy, V. Gopal, and R. Cutler. "DNSMOS P. 835: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors." in *Proc. ICASSP*, pp. 886-890, 2022.
- [8] Y. Yang, H. Taherian, V. A. Kalkhorani, and D.L. Wang, “Elevating robust ASR by decoupling multi-channel speaker separation and speech recognition,” in *Proc. ICASSP*, 2025, 5 pages.

If you encounter any issues, please do not hesitate to ask in the Class WeChat group.